MIL-C-38985A(USAF)
3 August 1979
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23 June 1971

MILITARY SPECIFICATION

CLIP-IN SUBASSEMBLY PWU-6B/C

1. SCOPE. This specification covers equipment designated as Clip-In Subassembly PWU-6B/C. With the Conversion Kit KMU-432/C or KMU-451/C installed the Clip-In Subassembly shall be a quick-loading device capable of retaining special weapons in a bomb bay without possibility of inadvertent release, and of releasing them without possibility of inadvertent retention.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extend specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods Of
MIL-E-7080	Electric Equipment, Aircraft, Selection and Installation Of
MIL-I-8500	Interchangeability and Replaceability Physical, of Component Parts for Aircraft and Missiles
MIL-A-8591	Airborne Stores and Associated Suspension Equipment; General Design Criteria For
MIL-R-28085	Rack, Bomb, Aircraft, Electric Type MB-3A
MIL-T-25821	Test Set, Armament Wiring System, Special Weapons AN/GWM-4
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Use Of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproof, with Appropriate Test Methods

FSC 1190

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DRAWINGS

Air Force

7613700-10	Clip-In Subassembly PWU-6B/C
7413600-30	Conversion Kit, Clip-In Subassembly KMU-451/C
7213000-30	Conversion Kit, Clip-In Subassembly KMU-432/C
7413700-10	Rack, Clip-In System MAU-175/A
7113118-10	Fixture, Calibrating

Sandia Laboratories

CD10001	B28FI Weapon

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or a directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION. -This specification makes provision for preproduction testing.

3.2 SELECTION OF MATERIALS. -Specifications and standards of all materials and parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

3.2.1 COMMERCIAL PARTS. -Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial untility parts like screws, bolts, nuts, cotter pins, etc., having suitable properities may be used provided:

a. They can be replaced by the standard parts (AN, MS or NAS) without aleration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractors's drawings.

3.2.2. STANDARD PARTS. -Except for use authorized in 3.2.1, AN, MS, and NAS standard parts shall be used where they suit the purpose. They shall be identifiied on the drawings by their part numbers.

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3.3 MATERIAL.-Materials shall be selected and processed for the various parts in accordance with the applicable specifications and standards as indicated on the drawings.

3.4 DESIGN AND CONSTRUCTION.-The PWU-6B/C Clip-In Subassembly shall be designed and constructed in strict conformance with drawing 7613700-10 and all subsidiary detail drawings, and shall effect support of the weapons, their release, protection against inadvertent release, monitoring and release circuitry, and connections to the weapons.

3.4.1 DISSIMILAR METALS IN CONTACT.-Dissimilar metals in contact as defined by MIL-STD-889 shall be protected from corrosion in accordance with the applicable provisions.

3.5 GENERAL STRUCTURAL CRITERIA

3.5.1 STRUCTURAL LOADS.-The Subassembly shall be capable of withstanding the structural loads specified in Table I(b) and of transmitting these loads to the clip-in points on the MAU-175/A Rack.

3.5.2 LIMIT ROAD.-Unless otherwise specified, the loads resulting from the loading conditions specified herein are limit loads.

3.5.3 YIELD FACTOR OF SAFETY.-The minimum yield factor of safety shall be 1.15. There shall be no permanent set as defined by MIL-A-8591.

3.5.4 ULTIMATE FACTOR OF SAFETY.-The minimum ultimate factor of safety shall be 1.50. Failure shall not occur when loads which are 1.50 times the load are applied. (Permanent set permissible).

3.6 PERFORMANCE.-The Subassembly shall be suitable for use in temperatures from -54°C (-65°F) to 71°C (160°F) and under adverse environmental conditions including sand and dust, salt water atmosphere and vibration, and shall meet the following performance requirements.

3.6.1 FUNCTIONAL

3.6.1.1 ELECTRICAL.-The electrical performance requirements shall be in accordance with 4.6.3.1

3.6.1.2 MECHANICAL LOCK MECHANISM.-The mechanical lock mechanism shall be capable of reversible actuation under limit loads. It shall be impossible to unlock after a release impulse has been applied to a rack in the locked condition. If a lock jams due to inadvertent release of a weapon, any other weapon shall be capable of being unlocked; except that, if a lower weapon lock is jammed, the lock of the weapon above shall not open.

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TABLE I

STRUCTURAL LOADS

(a) (POSITIVE SIGN CONVENTION SHOWN ON FIG. 1)

CONDITION	LIMIT LOAD FACTORS		
	Nx	Ny	Nz
1	-1.33	0	0*
2	0	+1.33	0*
3	0	0	3.67
4	-1.00	0	2.34

(b)

CONDITION	YIELD LOAD PER STORE		
	Px Lbs	Py Lbs	Pz Lbs
1	-3600	0	0*
2	0	+3600	0*
3	0	0	9920**
4	-2700	0	6330**

(c)

CONDITION	ULTIMATE LOAD PER STORE		
	Px Lbs	Py Lbs	Pz Lbs
1	-4700	0	0*
2	0	4700	0*
3	0	0	12950**
4	-3520	0	8250**

*Pz = 0 Indicates A Zero Gravity Condition

**Includes 2350 lb Weight of Store

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3.6.2 RELEASE.- The Subassembly shall be capable of 10 consecutive successful weapon releases from each MB-3A Rack.

3.6.3 HIGH TEMPERATURE.- The Subassembly shall be capable of satisfactory operation while exposed to a temperature of 71°C (160°F).

3.6.4 LOW TEMPERATURE.- The Subassembly shall be capable of satisfactory operation while exposed to a temperature of -54°C (-65°F).

3.6.5 HUMIDITY.- The Subassembly shall be capable of satisfactory operation while being subjected to the humidity test of this specification.

3.6.6 SALT SPRAY.- The Subassembly shall be capable of satisfactory operation after being exposed to a salt sea atmosphere and spray.

3.6.7 VIBRATION.- The Subassembly shall be capable of satisfactory operation after being subjected to the vibration tests of this specification.

3.6.8 SAND AND DUST.- The Subassembly shall be capable of satisfactory operation after extended exposure to blowing sand and dust.

3.7 ELECTRICAL REQUIREMENTS

3.7.1 Electrical equipment shall be installed in accordance with MIL-E-7080.

3.7.2 The Subassembly shall fulfill all operating requirements herein, when operating under any voltage from 18 to 30 volts dc.

3.8 INTERCHANGEABILITY.- All Subassemblies shall be interchangeable in accordance with the requirements of MIL-I-8500.

3.9 IDENTIFICATION OF PRODUCT.- Identification shall be in accordance with MIL-STD-130.

3.10 WORKMANSHIP.- The Subassembly, including all parts and accessories shall be fabricated in a manner that will insure that the criteria of appearance, fit, and adherence to specified tolerances and observed. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, plating, painting, riveting, and welding. All parts shall be free from burrs and sharp edges.

3.11 GOVERNMENT LOANED PROPERTY.- The following items of equipment shall be loaned to the contractor, upon his request, as Government loaned property, for the purpose of conducting the tests specified in section 4.

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Item No.	Part No.	Item Description	Quantity
1	7413700-10	Rack, Clip-In System MAU-175/A	As required (See 6.3)
2	330010-02	Power Supply T-128	1
3	111-100	Test Set, Special Weapon Aircraft Armament Wiring AN/GWM-4	1
4	7036933-10	Test Adapter, MX-9042/G	1
5	7113118	Fixture, Calibrating	2
6	69D13340-1	Trunnion	8
7	7213000-30	Conversion Kit, KMU-432/C	1

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFACTION OF TESTS. -The inspection and testing of the Subassembly shall be as follows:

a. Preproduction Tests.

b. Acceptance Tests.

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. -The preproduction test samples shall be representative of the production units. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.1.1 TEST REPORTS. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.2.2 PREPRODUCTION TESTS. Preproduction tests shall consist of all the following tests:

a. Individual

(1) Examination of Product
(2) Conformance to Drawings
(3) Functional Tests

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(a) Electrical
(b) Mechanical

b. Static Strength Tests

c. Release Test

d. Environmental Tests

(1) High Temperature
(2) Low Temperature
(3) Salt Spray
(4) Humidity
(5) Vibration
(6) Sand and Dust
(7) Temperature Shock

These tests are described under "4.6 TEST METHODS."

4.3 ACCEPTANCE TESTS. -Acceptance tests shall consist of the following individual tests described uner "4.6 TEST METHODS:"

a. Examination of Product

b. Conformance to Drawings

c. Functional Tests

4.4 TEST CONDITIONS

4.4.1 STANDARD CONDITIONS. -Standard conditions for conducting the tests shall be as follows:

a. Temperature: 25°C (70°F) plus or minus 10°C (18°F).

b. Relative Humidity: 90 percent or less.

4.4.2 TEST TEMPERATURE. -The entire subassembly shall be stabilized at the specified temperature prior to conducting any tests. Temperature stabilization shall be as defined in MIL-STD-810.

4.4.3 TEST MEASUREMENTS

4.4.3.1 INSTRUMENTS. -All measurements shall be made with instruments whose accuracy has been verified. Calibration records shall be maintained for submittal in the test report.

4.4.3.2 TOLERANCES. -The maximum allowable tolerances on test condition measurements shall be as follows:

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a. Vibration and shock amplitude - plus or minus 5 percent; frequency - plus or minus 2 percent.

NOTE: This tolerance is applicable only to the amplitude and frequency measuring instrument.

b. Applied load - plus or minus 5 percent in pounds.

4.4.3.3 REQUIRED MEASUREMENTS

a. Release time.

b. Release voltage and current.

c. Test load.

4.5 TEST EQUIPMENT REQUIRED. -Test equipment required for performance of testing is as follows:

a. Tester for electrical functional tests.

b. An 18-30 volt dc power source, T-128 (or equal).

c. Linear measuring equipment.

d. Suitable device for applying load to 58J46420 crank.

e. One MAU-175/A Rack.

f. Four simulated B-28FI weapons (shapes) reference Sandia Corp DWG No. CD10001 with attached lugs (lugs specified per MIL-A-8591) fore and aft 30 inches on center with provisions for loading per figure 1.

g. Four simualted B-28FI weapons with trunnions installed for release tests as shown in figure 2.

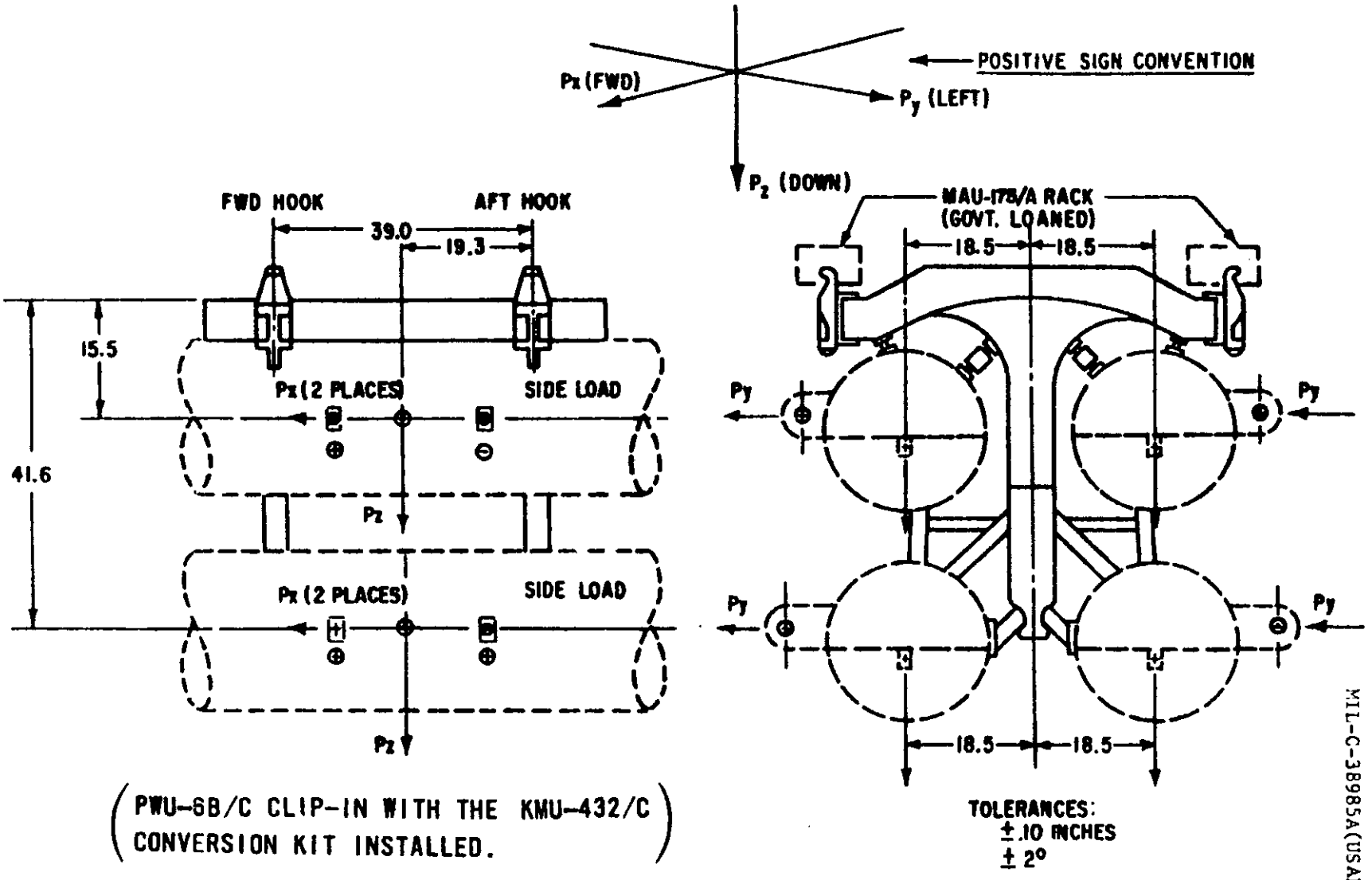
h. Test Set, Special Weapon Aircraft ARmament Wiring, AN/GWM-4.

i. Test Adapter, MX-9042/G.

4.6 TEST METHODS

4.6.1. EXAMINATION OF PRODUCT. -Each subassembly shall be given a complete visual examination to determine compliance with the requirements of this specification, including workmanship.

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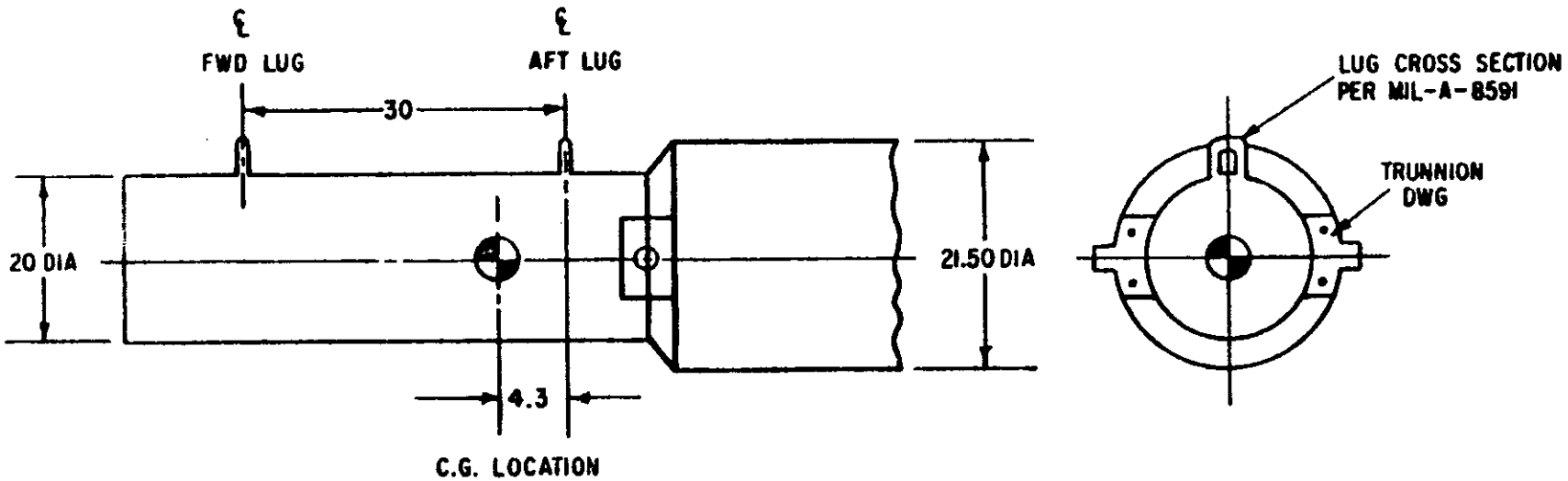
(PWU-6B/C CLIP-IN WITH THE KMU-432/C CONVERSION KIT INSTALLED.)

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FIGURE I. STATIC TEST SET-UP

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TOLERANCES:
± .10 INCHES

WEIGHT 2350 LBS.

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FIGURE 2. RELEASE TEST LOAD

4.6.2 CONFORMANCE TO DRAWINGS. Each Subassembly shall be inspected for strict conformance to the engineering drawings and all specifications and standards referenced therein.

4.6.3. FUNCTIONAL TESTS

4.6.3.1 ELECTRICAL SYSTEM FUNCTIONAL TESTS. Electrical system functional tests shall be made by use of the AN/GWM-4 Tester, MX-9042/G Test Adapter and T-128 Power Supply.

4.6.3.1.1 PRELIMINARY INSTRUCTIONS

a. Install the Fixture, Calibrating AF Dwg No. 7113118 to the fwd inboard plates holds the lower MB-3A racks.

b. Check that all power switches on T-128, MX-9042/G and AN/GWM-4 are in the OFF position and that all circuit breakers are open (pulled)

c. Using cable 7036793-50, connect P11 to J1 and P12 to J2 of AN/GWM-4

d. Using cable 7036793-50, connect P9 to J9 and P10 to J10 on MX-9042/G

e. Using cable 7036793-10, connect P2 to J2 and P4 to J4 on MX-9042/G

f. Using cable 7036793-10, connect P3 of cable to P3 of clip-in

g. Using cable 7036793-30, connect P6 to J6 and P8 to J8 on MX-9042/G

h. Using cable 7036793-30, connect P5 to P2 and P7 to P5 of clip-in

i. Using cable 7036792-30, connect P13 to J13 of MX-9042/G

j. Using cable 7036792-30, connect P15 of cable to T-128 power supply receptacle

4.6.3.1.2 CONTINUITY CHECKOUT PROCEDURE. Checkout procedures for the continuity and electrical functions shall be in accordance with Table II.

4.6.3.2 MECHANICAL LOCK SYSTEM FUNCTIONAL TEST. Functional testing of the mechanical lock system shall be in accordance with Table III.

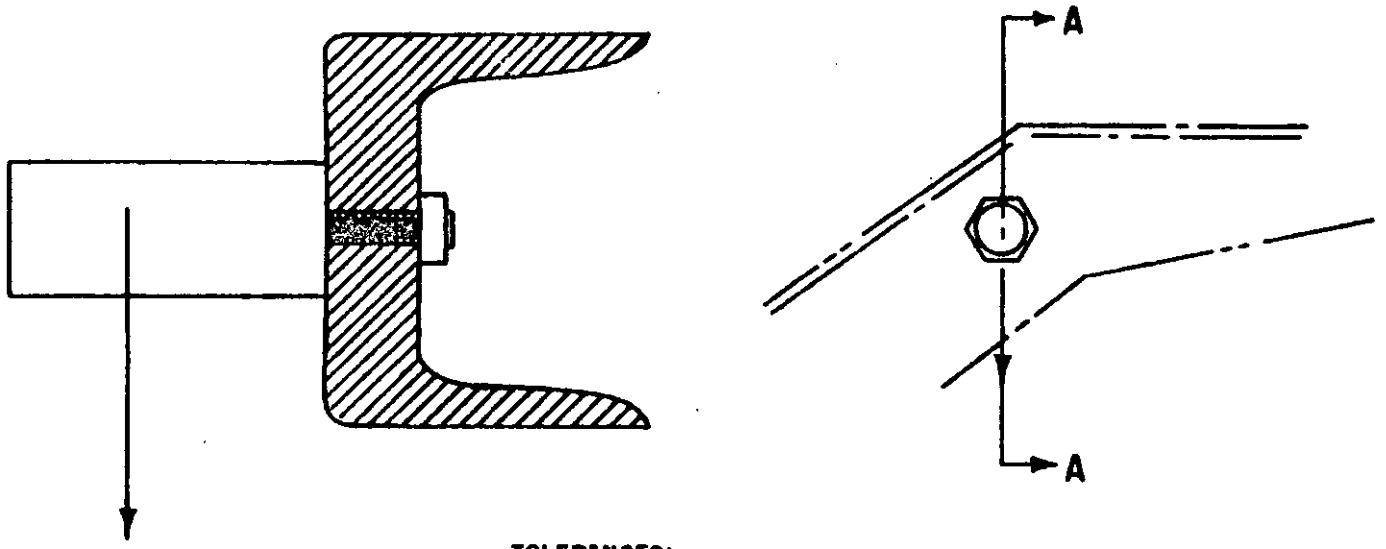
4.6.4 STATIC STRENGTH TESTS

4.6.4.1 STATIC TEST CONDITIONS. Static loads for the load condition are shown in Table I and Figures 3 and 4. Static load testing shall be conducted with the KMU-432/C conversion kit installed. The suggested method to apply loads obtained from Table I is shown in figure 1.

4.6.4.2 YIELD LOAD TEST. Yield loads (1.15 times the magnitude of the limit loads) as specified in Table I(b) and figures 3 and 4 shall be applied. The loads shall be applied for a period of not less than 10 seconds and there shall be no failure of any component as a result of this test.

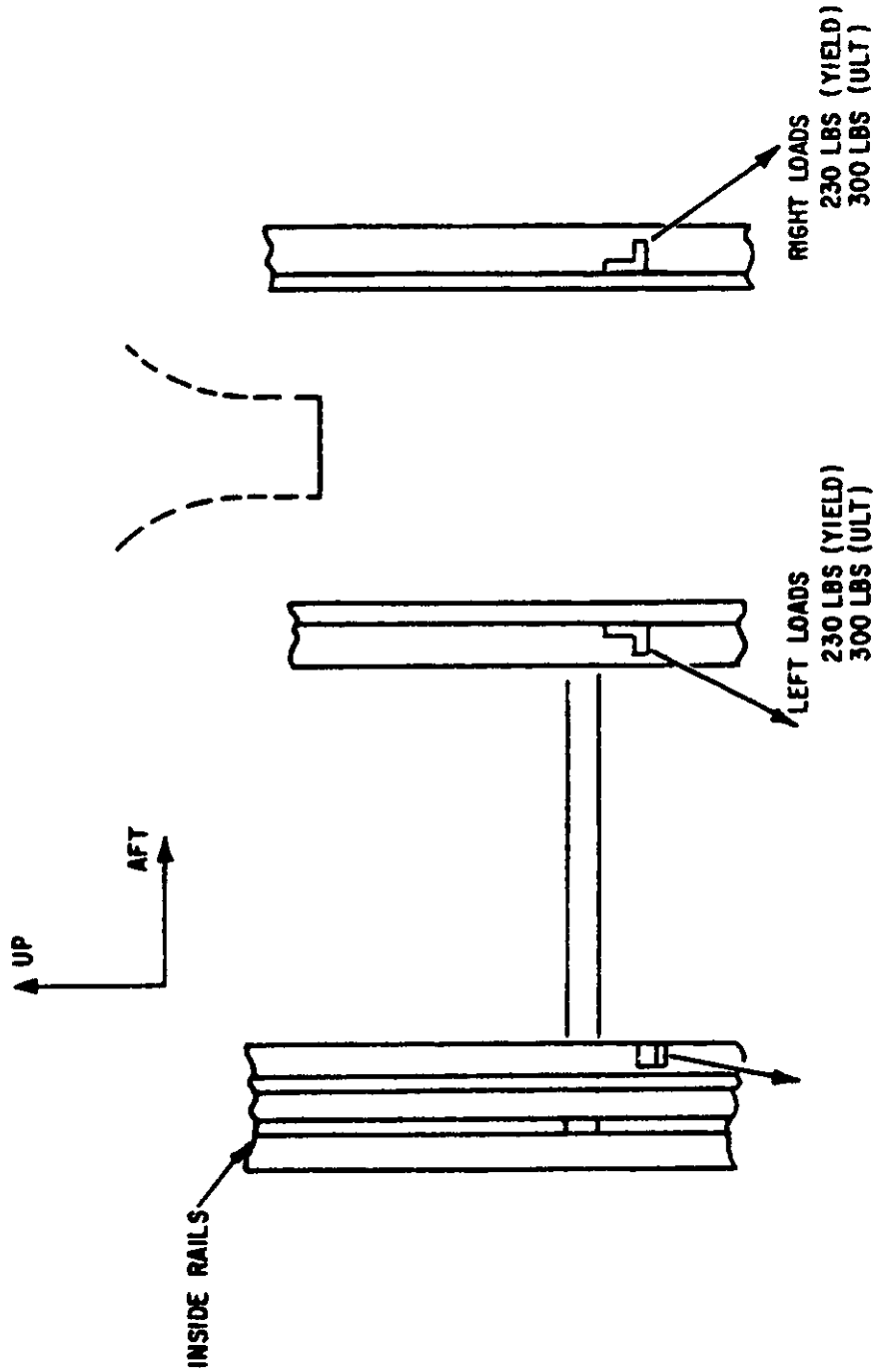
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NOTE: STRUCTURAL LOADS~UPPER PULLOUT BEAM LOAD SHOWN ACT SEPARATELY. EACH LOAD CAN ACT AT 10° PULLOFF ANGLE IN ANY DIRECTION FROM THE LINE OF ACTION SHOWN.

FIGURE 3. UPPER PULLOUT BEAM LOADS~ CONDITION I



NOTE: LOWER PULLOUT BEAM LOAD CONDITION. THE LOADS CAN ACT IN A 10° PULLOFF ANGLE IN ANY DIRECTION FROM THE LINE OF ACTION SHOWN.

FIGURE 4. LOWER PULLOUT BEAM LOADS

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TABLE II "CONTINUITY CHECK"

NOTE

The following steps (1 through 154) shall be performed in sequence

During tests disregard AN/GWM-4 FAULT lamp flickering POS "S" to POS "1".

Step	Procedure	Result
1	Turn on T-128 power supply and adjust to 28 vdc (±2) vdc	T-128 indicates 28 vdc (±2) vdc
2	Close circuit breaker on MX-9042/G	
3	Place POWER switch of MX-9042/G to ON	POWER lamp lights
4	On MX-9042/G, momentarily actuate CIRCUIT ADVANCE switch until "3" appears in POSITION window	"3" appears in POSITION window
5	Cock all bomb racks	Lockpins lock
6	Close all circuit breakers on AN/GWM-4	
7	On AN/GWM-4, set CIRCUIT SELECTOR switch to position 2	
8	On AN/GWM-4, set POWER switch to ON	POWER lamp lights
9	On AN/GWM-4, set switches as follows:	
	a. Set FUNCTION switch to CONTINUITY and/or INSULATION RESISTANCE and actuate START-OVERRIDE, AUTO-SCAN switch until "S" appears in both windows	"S" appears in CONTINUITY and INSULATION RESISTANCE windows
	b. FUNCTION switch to CONTINUITY	
	c. K2-RELAY-SIGNAL switch to K2-RELAY	
	d. POLARITY switch to (neg) -28V	
	e. METER switch to "1"	
	f. K1 RELAY switch to off (center position)	

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TABLE II

NOTE

SIGNAL indicator light should be OFF

If any circuit breaker should trip during the test, override to "S" position on AN/GWM-4 and reset breaker. Recheck procedures and continue with test.

Step	Procedure	Result
10	On AN/GWM-4, override to position 1	FAULT lamp lights
11	Actuate and hold drive crank to unlock position	FAULT lamp goes off
12	On AN/GWM-4, override to position 2	FAULT lamp remains off
13	Release drive crank to lock position	FAULT lamp lights
14	On AN/GWM-4, override to position 3	FAULT lamp remains on
15	Actuate and hold drive crank to unlocked position	FAULT lamp goes off
16	On AN/GWM-4, override to position 4	FAULT lamp remains off
17	Release drive crank to locked position	FAULT lamp lights
18	On AN/GWM-4, override to position 5	FAULT lamp remains on
19	Actuate and hold drive crank to unlocked position	FAULT lamp remains on
20	Manually trip lower left bomb rack	FAULT lamp goes off. Bomb rack uncocks
21	On AN/GWM-4, override to position 6	FAULT lamp lights
22	Manually trip lower right bomb rack	FAULT lamp goes off. Bomb rack uncocks
23	On AN/GWM-4, override to position 7	FAULT lamp lights
24	Manually trip upper left bomb rack	FAULT lamp goes off. Bomb rack uncocks
25	On AN/GWM-4, override to position 8	FAULT lamp lights
26	Manually trip upper right bomb rack	FAULT lamp goes off. Bomb rack uncocks.
27	On AN/GWM-4, override to position 9	FAULT lamp lights

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TABLE II

Step	Procedure	Result
28	Cock all bomb racks	FAULT lamp remains on
29	Release drive crank to locked position	FAULT lamp goes off
30	On AN/GWM-4, override to position 10	FAULT lamp remains off
31	Actuate and hold drive crank to unlocked position	FAULT lamp lights
32	On AN/GWM-4, override to position 11	FAULT lamp remains on
33	Release drive crank to locked position	FAULT lamp goes off
34	On AN/GWM-4, override to position 12	FAULT lamp remains off
35	Actuate and hold drive crank to unlocked position	FAULT lamp lights
36	On AN/GWM-4, override to position 13	FAULT lamp goes off
37	On AN/GWM-4, override to position 14	FAULT lamp remains off
38	On AN/GWM-4, override to position 15	FAULT lamp lights
	CAUTION Do not hold RELEASE/SALVO switch on MX-9042/G in position "1" or "2" more than three seconds. Only a momentary action of switch is necessary	
39	On MX-9042/G, momentarily actuate RELEASE/SALVO switch to position 1	Lower right bomb rack releases
40	On MX-9042/G, momentarily actuate RELEASE/SALVO switch to position 2	Lower left bomb rack releases
41	On MX-9042/G, momentarily actuate RELEASE/SALVO switch to position 1	Upper right bomb rack releases
42	On MX-9042/G, momentarily actuate RELEASE/SALVO switch to position 2	Upper left bomb rack releases
43	Cock all bomb racks	
44	Release drive crank to locked position	
45	Install shorting plug, Part No. 7036791-10, on (LL) of RYK-2/A24T-1 relay box	

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TABLE II

Step	Procedure	Result
46.	Actuate and hold crank to unlocked position	
47	On AN/GWM-4, actuate and hold K1 RELAY switch in on (down) position	FAULT lamp goes off
	CAUTION Do not hold RELEASE/SALVO switch on MX-9042/G in position 2 more than three seconds. Only momentary action of switch is necessary	
48	On MX-9042/G, momentarily actuate RELEASE/SALVO switch to position 2	All bomb racks release
49	On AN/GWM-4, release K1 RELAY switch	FAULT lamp lights
50	Release drive crank to locked position	FAULT lamp remains on
51	On AN/GWM-4, override to position 16	FAULT lamp remains on
52	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
53	Release K1 RELAY switch	FAULT lamp lights
54	On AN/GWM-4, override to position 17	FAULT lamp remains on
55	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
56	Release K1 RELAY switch	FAULT lamp lights
57	On AN/GWM-4, set K2 RELAY-SIGNAL switch to off (center) position	FAULT remains on
58	Install shorting plug on UL	FAULT lamp remains on
59	On AN/GWM-4, override thru "S" to position 15	FAULT lamp remains on, except in "S" position
60	On AN/GWM-4, set K2 RELAY-SIGNAL switch to K2 RELAY	FAULT lamp remains on
61	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
62	On AN/GWM-4, release K1 RELAY switch	FAULT lamp lights

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TABLE II

Step	Procedure	Result
63	On AN/GWM-4, override to position 16	FAULT lamp remains on
64	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
65	Release K1 RELAY switch	FAULT lamp lights
66	On AN/GWM-4, override to position 17	FAULT lamp remains on
67	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
68	Release K1 RELAY switch	FAULT lamp lights
69	On AN/GWM-4, set K2 RELAY-SIGNAL switch to off (center) position	FAULT lamp remains on
70	Install shorting plug on UR	FAULT lamp remains on
71	On AN/GWM-4, override thru "S" to position 15	FAULT lamp remains on Except in "S" position
72	On AN/GWM-4, set K2 RELAY-SIGNAL switch to K2 RELAY	FAULT lamp remains on
73	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
74	Release K1 RELAY switch	FAULT lamp lights
75	On AN/GWM-4, override to position 16	FAULT lamp remains on
76	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
77	Release K1 RELAY switch	FAULT lamp lights
78	On AN/GWM-4, override to position 17	FAULT lamp remains on
79	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
80	Release K1 RELAY switch	FAULT lamp lights
81	On AN/GWM-4, set K2 RELAY-SIGNAL switch to off (center) position	FAULT lamp remains on

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TABLE II

Step	Procedure	Result
82	Install shorting plug on LR	FAULT lamp remains on
83	On AN/GWM-4, override thru "S" to position 15	FAULT lamp remains on, except in "S" position
84	On AN/GWM-4, set K2 RELAY-SIGNAL switch to K2 RELAY	FAULT lamp remains on
85	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
86	Release K1 RELAY switch	FAULT lamp lights
87	On AN/GWM-4, override to position 16	FAULT lamp remains on
88	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
89	Release K1 RELAY switch	FAULT lamp lights
90	On AN/GWM-4, override to position 17	FAULT lamp remains on
91	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
92	Release K1 RELAY switch	FAULT lamp lights
93	On AN/GWM-4, actuate K2 RELAY SIGNAL switch to off (center) position	FAULT lamp remains on
94	On AN/GWM-4, override to "S" position	FAULT lamp goes off
95	On MX09042/G, momentarily actuate CIRCUIT-ADVANCE switch until "1" appears in POSITION window	FAULT lamp remains off
96	On AN/GWM-4, override to Position 1	FAULT lamp remains off
97	On AN/GWM-4, auto-scan to position 3	FAULT lamp lights
98	On AN/GWM-4, override to position 4	FAULT lamp goes off
99	On AN/GWM-4, auto-scan to position 7	FAULT lamp lights
100	On AN/GWM-4, override to position 8	FAULT lamp remains on
101	On AN/GWM-4, override to position 9	FAULT lamp remains on
102	On AN/GWM-4, override to position 10	FAULT lamp goes off

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TABLE II

Step	Procedure	Result
103	On AN/GWM-4, override to position 11	FAULT lamp lights
104	On AN/GWM-4, override to position 12	FAULT lamp goes off
105	On AN/GWM-4, override to position 13	FAULT lamp lights
106	On AN/GWM-4, override to position 14	FAULT lamp remains on
107	On An/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
108	Release K1 RELAY switch	FAULT lamp lights
109	On AN/GWM-4, override to "S" position	FAULT lamp remains on at positions 15 thru 17 - FAULT lamp goes off at 18 and remains off
110	Install shorting plug on UR	FAULT lamp remains off
111	On AN/GWM-4, override to position 1	FAULT lamp remains off
112	On AN/GWM-4, auto-scan to position 3	FAULT lamp lights
113	On AN/GWM-4, override to position 4	FAULT lamp goes off
114	On AN/GWM-4, auto-scan to position 7	FAULT lamp lights
115	On AN/GWM-4, override to position 8	FAULT lamp remains on
116	On AN/GWM-4, override to position 9	FAULT lamp remains on
117	On AN/GWM-4, override to position 10	FAULT lamp goes off
118	On AN/GWM-4, override to position 11	FAULT lamp lights
119	On AN/GWM-4, override to position 12	FAULT lamp goes off
120	On AN/GWM-4, override to position 13	FAULT lamp lights
121	On AN/GWM-4, override to position 14	FAULT lamp remains on
122	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
123	Release K1 RELAY switch	FAULT lamp lights
124	On AN/GWM-4, override to "S" position	FAULT lamp remains on at position 15 thru 17 - FAULT lamp goes off at 18 and remains off

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TABLE II

Step	Procedure	Result
125	Install shorting plug on UL	FAULT lamp remains off
126	On AN/GWM-4, override to position 1	FAULT lamp remains off
127	On AN/GWM-4, auto-scan to position 7	FAULT lamp lights
128	On AN/GWM-4, override to position 8	FAULT lamp remains on
129	On AN/GWM-4, override to position 9	FAULT lamp remains on
130	On AN/GWM-4, override to position 10	FAULT lamp goes off
131	On AN/GWM-4, override to position 11	FAULT lamp lights
132	On AN/GWM-4, override to position 12	FAULT lamp remains on
133	On AN/GWM-4, override to position 13	FAULT lamp remains on
134	On AN/GWM-4, override to position 14	FAULT lamp remains on
135	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
136	Release K1 RELAY switch	FAULT lamp lights
137	On AN/GWM-4, override to "S" position	FAULT lamp remains on at positions 15 thru 17 - FAULT lamp goes off at 18 and remains off
138	Install shorting plug on UL	FAULT lamp remains off
139	On AN/GWM-4, override to position 1	FAULT lamp remains off
140	On AN/GWM-4, auto-scan to position 7	FAULT lamp lights
141	On AN/GWM-4, override to position 8	FAULT lamp remains on
142	On AN/GWM-4, override to position 9	FAULT lamp remains on
143	On AN/GWM-4, override to position 10	FAULT lamp goes off
144	On AN/GWM-4, override to position 11	FAULT lamp lights
145	On AN/GWM-4, override to position 12	FAULT lamp remains on
146	On AN/GWM-4, override to position 13	FAULT lamp remains on
147	On AN/GWM-4, override to position 14	FAULT lamp remains on

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TABLE II

Step	Procedure	Result
148	On AN/GWM-4, actuate and hold K1 RELAY switch to on (down) position	FAULT lamp goes off
149	Release K1 RELAY switch	FAULT lamp lights
150	On AN/GWM-4, override to "S" position	FAULT lamp remains on at positions 15 thru 17 - FAULT lamp goes off at 18 and remains off
151	Remove and stow shorting plug	
152	Set AN/GWM-4 and MX-9042/G POWER switches to OFF and pull all circuit breakers	
153	Turn T-128 power supply to OFF	
154	Disconnect all equipment and stow cables	

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TABLE III

MECHANICAL LOCK SYSTEM FUNCTIONAL TEST		
STEP	OPERATION	RESPONSE
1. a.	Apply a force of 85 pounds to the outboard end of the 58J46420 drive crank	(1) The crank shall move to the fully unlocked position. (2) All lock pins shall be fully retracted.
b.	Release the drive crank	The lock pins shall return to the fully locked position.
2. a.	With a test load as defined in 4.5g on each MB-3A Rack, and the lock pins fully inserted, actuate the manual release on all racks.	The hooks shall not open.
b.	Apply a downward load of 210 pounds to the roller on the outboard end of the 58J46420 drive crank.	The lock pins shall not retract and the loads shall not release.
c.	Release the down load on the drive crank, relieve the test load and re-cock all racks.	
3. a.	With a test load as defined in 4.5g on each MB-3A Racks and the lock pins fully inserted, pull the manual release on the lower right hand rack.	The load shall not release.
b.	Apply a down load of 210 pounds to the roller on the outboard end of the 58J46420 drive crank.	(1) The lock pin in the two left hand rack shall retract.

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TABLE III

MECHANICAL LOCK SYSTEM FUNCTIONAL TEST		
STEP	OPERATIONAL	RESPONSE
3. b. (Cont'd)		(2) The lock pin in the two right hand racks shall not retract.
c.	Release the down load on the drive crank, relieve the test load and re-cock the lower right hand rack.	
4. a.	With a test load as defined in 4.5g on each MB-3A Racks and the pins fully inserted, pull the manual release on the lower left hand rack.	The load shall not release
b.	Apply a down load of 210 pounds to the roller on the outboard end of the 58J46420 drive crank.	(1) The lock pin in the two right hand racks shall retract. (2) The lock pins in the left hand racks shall not retract.
c.	Release the down load on the drive crank, relieve the test load and re-cock the lower left hand rack.	
5. a.	With a test load as defined in 4.5g on each MB-3A Rack, and the lock pins fully inserted, pull the manual release on the upper right hand rack.	The load shall not release.
b.	Apply a down load of 210 pounds to the roller on outboard end of the 58J46420 drive crank.	(1) The lock pin in the upper right hand rack shall not retract.

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TABLE III

MECHANICAL LOCK SYSTEM FUNCTIONAL TEST		
STEP	OPERATION	RESPONSE
5. b.	(Cont'd)	(2) The lock pins in the left hand racks and in the lower right hand rack shall retract.
c.	Release the down load on the drive crank, relieve the test load and re-cock the upper right hand rack.	
6. a.	With a test load as defined in 4.5g on each MB-3A Rack, and the lock pins fully inserted, pull the manual release on the upper left hand rack.	The load shall not release.
b.	Apply a down load of 210 pounds to the roller on outboard end of the 58J46420 drive crank.	(1) The lock pin in the upper left hand rack shall not retract. (2) The lock pins in the right hand racks and in the lower left hand rack shall retract.
c.	Release the down load on the drive crank, relieve the test load and re-cock the upper left hand rack.	
7. a.	With test load as defined in 4.5g on each MB-3A Rack, move the roller on the outboard end of the 58J46420 drive crank to the full down position and hold. Pull the manual release on the lower right hand rack.	The hooks shall open and the load shall release.

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TABLE III

MECHANICAL LOCK SYSTEM FUNCTIONAL TEST		
STEP	OPERATION	RESPONSE
7. (Cont'd) b.	Release the drive crank.	The lock pins in the left hand racks and in the upper right hand rack shall return to the locked position.
c.	Move the drive crank to the down position and hold. Pull the manual release on the upper right hand rack.	The hooks shall open and the load shall release.
d.	Release the drive crank	The lock pins in the left hand racks shall return to the locked position.
e.	Move the drive crank to the full down position and hold. Pull the manual release on the lower left hand rack.	The hooks shall open and load shall release.
f.	Release the drive crank.	The lock pin in the upper left hand rack shall return to the locked position.
g.	Move the drive crank to the down position, and hold. Pull the manual release on the upper left hand rack.	The hooks shall open and the load shall release.
h.	Release the drive crank.	

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TABLE IV

RELEASE TEST LOADS

V(a) (POSITIVE SIGN CONVENTION SHOWN IN FIG. 1)

	LOAD FACTORS		
COND	Nx	Ny	Nz
1	+.44	+.55	+3.30
2	-.44	-.55	+3.30
3	-.44	-.55	+ .75

V(b) EXTERNAL LOADS APPLIED

	YIELD LOADS PER STORE		
COND	Px Lbs	Py Lbs	Pz Lbs
1	1190	1485	6590
2	-1190	-1485	6590
3	-1190	-1485	- 325

V(c) EXTERNAL LOADS APPLIED

	ULTIMATE LOAD PER STORE		
COND	Px Lbs	Py Lbs	Pz Lbs
1	1550	1940	9270
2	-1550	-1940	300

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4.6.4.3 ULTIMATE LOAD TESTS. Ultimate loads (1.50 times the magnitude of specified limit loads) as specified in Table I (c) and figures 3 and 4 shall be applied. The loads shall be applied for a period of not less than 10 seconds and there shall be no failures of any component as a result of this test.

4.6.5 RELEASE TESTS

4.6.5.1 Free Fall Release Tests. The free fall release tests shall consist of two consecutive successful drops of the simulated B-28FI Bomb (ref figure 2) from each MB-3A Racks with the subassembly installed at positions with the nose-up pitch angles of 0, 3, 5, 7 1/2, and 10 degrees. Operation of the Mechanical Lock System shall be checked in accordance with Steps 1 and 2 of Table III. Failure to release shall constitute an unsuccessful drop. At the conclusion of the drops, all load carrying members shall be inspected for indications of damage resulting from bomb drop, the occurrence of which shall be cause for rejection.

4.6.5.2 Loaded Release Test. The loaded release test shall be performed with the subassembly loaded with simulated B28FI stores reference figure 2, released from each MB-3A racks with applied loads specified in Table IV. Failure to release shall constitute an unsuccessful drop. At the conclusion of the drops, all load carrying members shall be inspected for indicators of damage resulting from bomb drop, the occurrence of which shall be cause for rejection.

4.6.6 ENVIRONMENTAL TESTS

4.6.6.1 GENERAL. The Subassembly shall be loaded with test loads as defined in 4.5a while being subjected to the specified environmental tests. Unless otherwise specified in the following detailed test procedures, operation specified in MIL-STD-810 shall consist of one cycle of the release tests of this specification.

4.6.6.2 HIGH TEMPERATURE. The Subassembly shall be tested in accordance with the requirements of procedure II of MIL-STD-810.

4.6.6.3 LOW TEMPERATURE. The Subassembly shall be tested in accordance with the requirements of procedure II of MIL-STD-810.

4.6.6.4 HUMIDITY. The Subassembly shall be tested in accordance with the requirements of procedure I of MIL-STD-810.

4.6.6.5 SALT SPRAY. The Subassembly shall be tested in accordance with the requirements of procedure I of MIL-STD-810.

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4.6.6.6 VIBRATION. The subassembly shall be tested in accordance with procedure XII of MIL-STD-810, except that the range of frequencies and vibratory accelerations shown in figure 5 shall apply in lieu of Table I of MIL-STD-810. The test shall be conducted in two parts.

a. Each of the MB-3A Racks shall be released at the completion of the resonance frequency vibration test, while the subassembly is being vibrated at resonanat frequency.

b. The subassembly shall be tested in accordance with steps 1 and 7 of Table III subsequent to the completion of the vibration tests.

At the completion of the test, the electrical system shall meet the test requirements of the Electrical System Functional Tests of this specification.

4.6.6.7 SAND AND DUST. One subassembly, with test loads (4.5g) installed, shall be subjected to tests in accordance with Procedure I, parts I and II of MIL-STD-810. Sand and dust velocity shall be 2500±500 feet per minute. At the end of exposure period, the equipments shall be removed from the test chamber, and allowed to cool to room temperature. The following test shall then be conducted:

Mechanical Lock System Functional Test, Table III, steps 1 and 7.

4.6.6.8 TEMPERATURE SHOCK. The subassembly, with test loads installed shall be placed in a test chamber and stabilized at -54°C (-65°F). The subassembly shall then be exposed to an atmosphere having a temperature of 73°C (165°F) and a relative humidity of 90 percent for a period of 4 hours. Retaining all condensation practicable, the subassembly shall be exposed to an atmosphere of -54°C (-65°F) for a period of 4 hours. There shall be three such cycles from 73°C (165°F) to -54°C (-65°F). At the conclusion of this specification, either while still in the chamber, or within 5 minutes after removal from the chamber.

4.6.7 BASIS FOR REJECTION

a. Individual units failing to meet their respective tests shall be rejected. Rejected units either shall be replaced or shall be reworked to correct the defects, after which all specified tests shall be repeated. Rejected units shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and meaures taken to overcome defects.

b. All parts, speciments or assemblies destroyed in making tests required by this specification shall be in addition to the quantity specified

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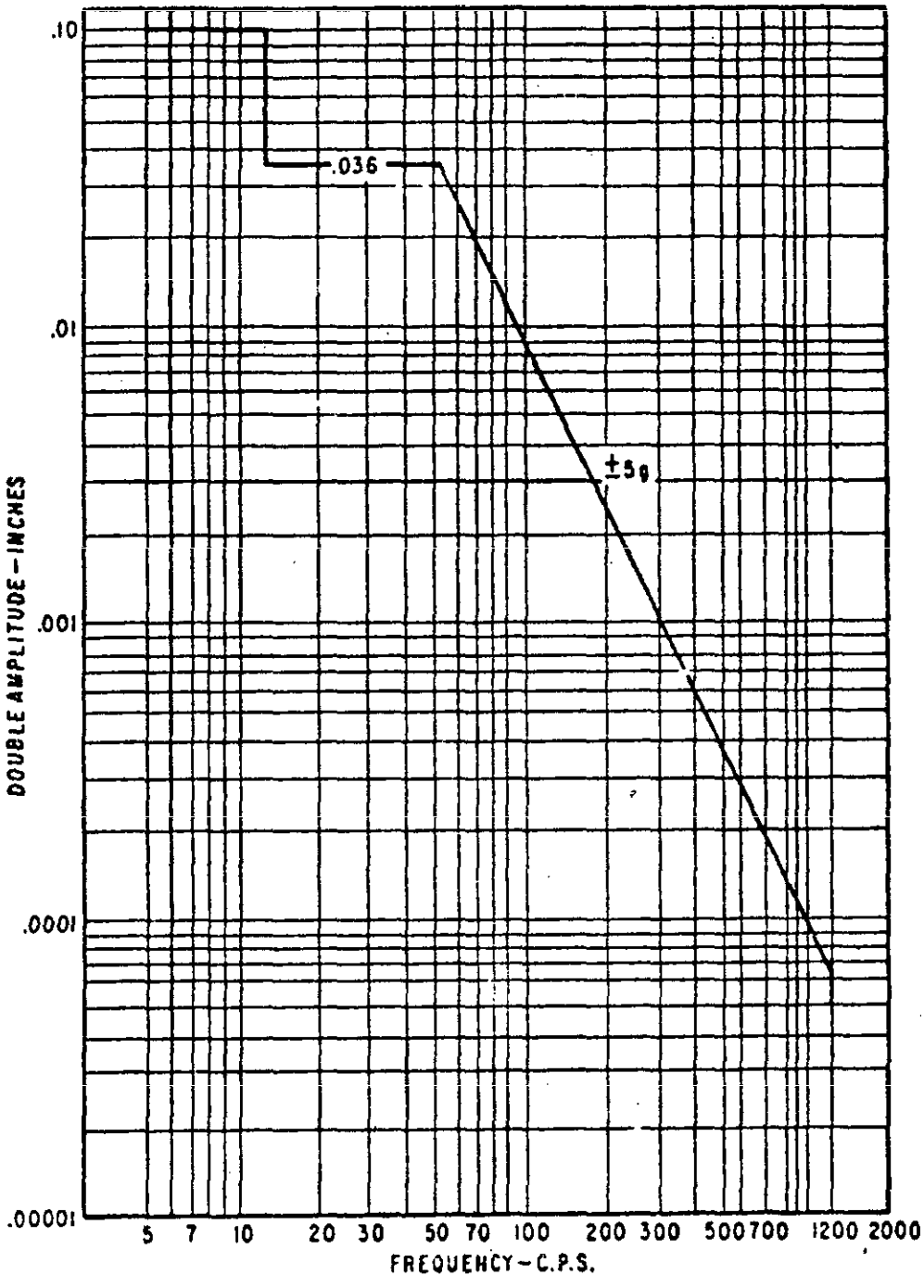


FIGURE 5. RANGE CURVE FOR VIBRATION TEST

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in the order, and shall be furnished to the procuring activity without increasing the cost of the contract. Accepted units shall have such acceptance marking on the unit as specified by the procuring activity.

c. After any test, the occurrence of any of the following conditions shall be sufficient cause for rejection:

Failure to perform within required tolerances.

Failure of structure, whether by deformation, rupture, or misalignment.

Dirty or pitted electrical contact points.

Dirt, metal chips, filings, solder or burns on any surface.

Parts not properly finished.

Loose connections, badly soldered joints or frayed ends of wires.

5. PREPARATION FOR DELIVERY

5.1 LEVELS OF PRESERVATION AND PACKAGING

5.1.1 PRESERVATION. Preservation for all shipment levels shall be in accordance with method I of MIL-P-116.

5.1.2 PACKAGING AND PACKING. The quantity of Subassemblies per container shall be specified by the procuring activity.

5.1.2.1 LEVELS A AND B. The Subassemblies shall be physically protected in accordance with MIL-STD-1186 and packed in a box conforming to PPP-B-601 or PPP-B-621.

5.1.2.2 LEVEL C. The Subassemblies for level C shipment shall be packed in exterior type containers in such a manner as to insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules, or the regulations of other common carriers as applicable.

5.2 MARKING FOR SHIPMENT AND STORAGE. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be:

CLIP-IN SUBASSEMBLY PWU-6B/C

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6. NOTES

6.1 INTENDED USE. The Subassembly with a conversion kit installed is intended for carrying special weapons in high-performance bomber aircraft and maintaining readiness during storage.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. The title, number, and data of this specification.

b. Levels of preservation, packaging and packing.

c. Whether preproduction tests will be conducted by the procuring activity.

6.2.1 PREPRODUCTION TEST SAMPLES. It is expected that the contract or purchase order will specify that a minimum of two Subassemblies representative of the production equipment will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order, should specify the point of inspection for these tests.

6.3 GOVERNMENT LOANED PROPERTY. The quantity of Government loaned property may be varied according to the production rate.

6.4 PRODUCTION DELIVERY. No production items should be delivered prior to approval of the preproduction samples.

6.5 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order to prove, prior to his starting production, that the Subassembly meet the requirements of this specification. Any production prior to the approval of the preproduction samples shall be at the contractor's risk.

b. ACCEPTANCE TESTS. Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification and to determine that the units are equivalent in performance and construction to the approved preproduction samples.

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